Case Report

A Large Choroidal Melanoma In An Indian Male: A Case Report With Clinicopathological Correlation

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Abstract

Background: Choroidal melanoma is the most common primary intraocular malignancy in adults. It arises from melanocytes of the uveal tract and is considerably rarer in Asian populations compared to Western cohorts. Because early symptoms may be absent or nonspecific, diagnosis can be delayed, particularly in regions where the disease is uncommon. Multimodal ocular imaging, supplemented by histopathological evaluation when tissue is available, is critical for establishing the diagnosis and guiding management. We report a large choroidal melanoma in an Indian male and describe its clinical, radiological, pathological, and management features.

Case report: A 50-year-old man presented with a two-month history of progressive, painless vision loss in the right eye. Fundus examination showed a large, elevated, pigmented choroidal mass involving the posterior segment. B-scan ultrasonography demonstrated a dome-shaped intraocular lesion with low-to-medium internal acoustic reflectivity. Magnetic resonance imaging revealed a well-defined postero-medial intraocular mass that was hyperintense on T1-weighted images and hypointense on T2-weighted images. Systemic screening did not identify distant metastasis. Because of the tumor's size and poor visual potential, right-eye enucleation was performed. Histopathological examination confirmed spindle-cell choroidal melanoma measuring 19 mm in basal diameter and 9 mm in thickness, without optic nerve or extraocular invasion.

Conclusion: Although rare in Asian populations, choroidal melanoma should be considered in middle-aged or older individuals with unexplained, progressive visual loss and a pigmented intraocular mass. This case underscores that for large tumors, enucleation remains the definitive treatment, and it highlights the essential need for structured long-term metastatic surveillance—with particular emphasis on hepatic monitoring—to optimize outcomes.

Keywords: Choroidal melanoma; Uveal melanoma; Magnetic resonance imaging; Histopathology; Enucleation; Metastasis; Liver surveillance; Ocular oncology

Introduction:

Uveal melanoma is a malignant tumor arising from melanocytes within the uveal tract, which includes the iris, ciliary body, and choroid [1]. Although it constitutes a small proportion of all melanomas, it is the most common primary intraocular malignancy in adults [2]. The choroid is the most frequently involved site, accounting for approximately 85–90% of cases [3].

The epidemiology of uveal melanoma varies markedly across ethnic groups. Reported incidence is highest among Caucasian populations, generally around 5–8 cases per million per year, whereas Asian and African populations have substantially lower rates [1,4]. This difference has been attributed to a combination of phenotypic and genetic factors, including skin pigmentation and iris color [5,6].

Most patients are diagnosed between the fifth and eighth decades of life [5]. Small tumors may remain clinically silent, whereas larger or more centrally located lesions may cause blurred vision, photopsia, floaters, visual-field defects, or metamorphopsia, depending on the tumor's location, size, and associated retinal detachment [7].

Unlike many other malignancies, choroidal melanoma typically spreads through the bloodstream rather than lymphatic channels. The liver is the most frequent site of metastasis, followed by the lungs and bones [6,8]. Once distant metastasis develops, the prognosis is poor, with a median survival of less than one year [1,7].

Diagnosis is usually based on clinical examination supported by ocular ultrasonography, fundus imaging, and magnetic resonance imaging. B-scan ultrasonography is

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a useful and accessible tool for assessing tumor configuration and internal reflectivity, while MRI helps define tumor extent, signal characteristics, and possible extra s cleral spread. Histopathological examination provides definitive tissue confirmation in enucleated or biopsied specimens [7,9]. We report a case of a large choroidal melanoma in an Indian male, with clinicopathological correlation and a review of the relevant literature.

Case Report:

Clinical presentation

A 50-year-old man attended the Department of Ophthalmology, Index Medical College Hospital and Research Centre, Indore, with gradual, painless loss of vision in the right eye for two months. He did not report redness, watering, flashes, headache, fever, trauma, drug abuse, or any known systemic illness.

Ophthalmic examination

Best-corrected visual assessment showed perception of light with inaccurate projection of rays in the right eye. The left eye had a distance visual acuity of 6/12, improving to 6/6p with correction. Near vision in the left eye improved from N8 to N6 with a +1.25 D spherical add. Both corneas and conjunctivae were clear. The anterior chamber depth was normal in both eyes, corresponding to Van Herick grade 4. Intraocular pressure by Goldmann applanation tonometry was 14 mmHg in each eye. Pupils were central, round, and reactive to light. Both crystalline lenses were clear, with no evidence of cataract. Lid position and extraocular movements were normal, and ocular motility was full in all directions of gaze.

Fundus examination of the right eye revealed a large, dome-shaped, elevated, pigmented choroidal mass arising from the nasal choroid and extending posteriorly to overshadow the macular region, with a dull foveal reflex. Areas of shallow retinal detachment were noted adjacent to the lesion. No vitreous hemorrhage or significant vitritis was observed. The left fundus showed clear media, a healthy optic disc with well-defined margins, normal retinal vessels, a bright foveal reflex, and no abnormal pigmentation or elevation. These findings were consistent with a large pigmented intraocular mass, clinically suggestive of choroidal melanoma.

Slit-lamp fundus evaluation demonstrated a reddish-brown intraocular mass in the posterior segment (Figure 1). B-scan ultrasonography showed a dome-shaped intraocular mass occupying the posterior segment, with low-to-medium internal reflectivity, supporting the diagnosis of choroidal melanoma (Figure 2).

Table 1. Ophthalmic examination findings

Parameter	Right eye	Left eye
Distance vision	Perception of light with inaccurate projection of rays	6/12 with -1.00 cyl at 150° → 6/6p
Near vision	—	N8 with +1.25 sph add → N6
Cornea	Clear	Clear
Conjunctiva	Clear	Clear
Anterior chamber (Van Herick grading)	Grade 4	Grade 4
Intraocular pressure (GAT)	14 mmHg	14 mmHg
Pupil	RAPD +	CCRL+
Lens	Clear	Clear
Lids	Normal	Normal

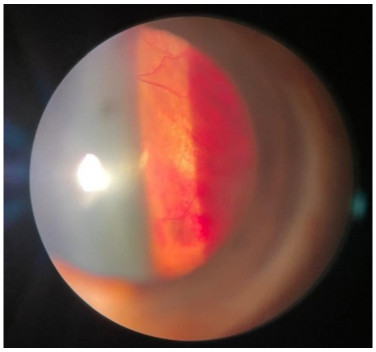

Figure 1. Slit-lamp fundus photograph showing a dome-shaped, reddish-brown choroidal mass in the posterior segment of the right eye.

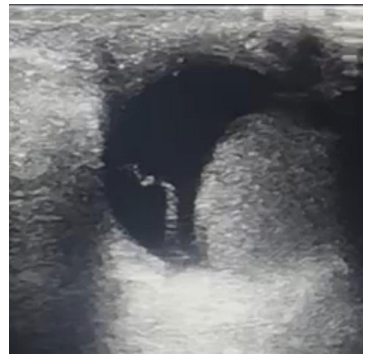

Figure 2. B-scan ultrasonography demonstrating a dome-shaped choroidal mass occupying the posterior segment of the right eye.

Imaging findings

Contrast-enhanced MRI of the orbits identified a well-circumscribed, dome-shaped lesion in the postero-medial aspect of the right globe, measuring 1.8 × 1.5 × 1.9 cm. Relative to the vitreous, the lesion was hyperintense on T1-weighted images and hypointense on T2-weighted images (Figure 3). These signal characteristics are typical of melanotic melanoma and reflect the paramagnetic properties of melanin and associated free radicals [8,9]. No optic nerve invasion or extraocular extension was detected.

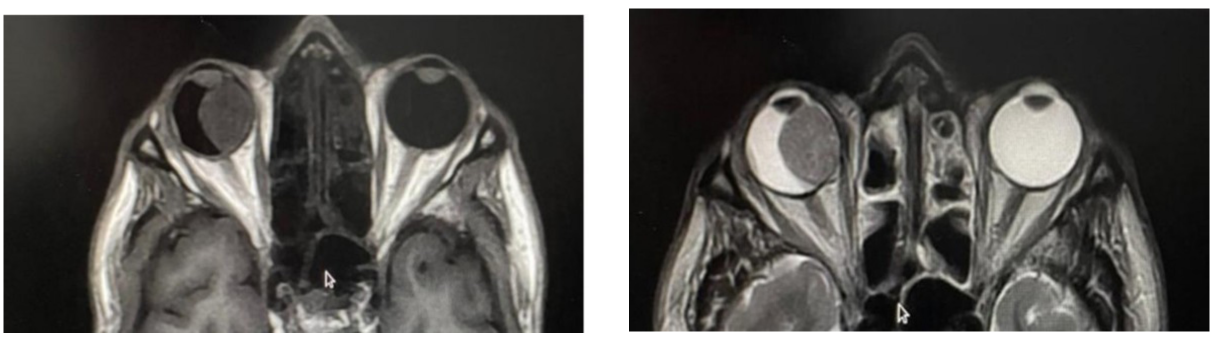

Figure 3. Contrast-enhanced MRI of the orbits showing a well-defined, dome-shaped intraocular lesion in the postero-medial right globe. The lesion is hyperintense on T1-weighted imaging and hypointense on T2-weighted imaging, consistent with choroidal melanoma.

Systemic evaluation and management

Systemic evaluation, including chest radiography, abdominal ultrasonography and general clinical examination, did not reveal metastatic disease. In view of the large tumor size, posterior segment involvement and poor visual prognosis, right-eye enucleation was performed under general anesthesia. The enucleated globe was submitted for histopathological examination.

Histopathological findings

Gross examination showed a dome-shaped, deeply pigmented, endophytic intraocular tumor arising from the choroid. The tumor measured approximately 1.9 cm (19 mm) in largest basal diameter and 9 mm in thickness (Figure 4). It appeared to arise from the nasal choroid and extended posteriorly, overshadowing the macular region clinically, with no obvious scleral or optic nerve invasion on gross assessment.

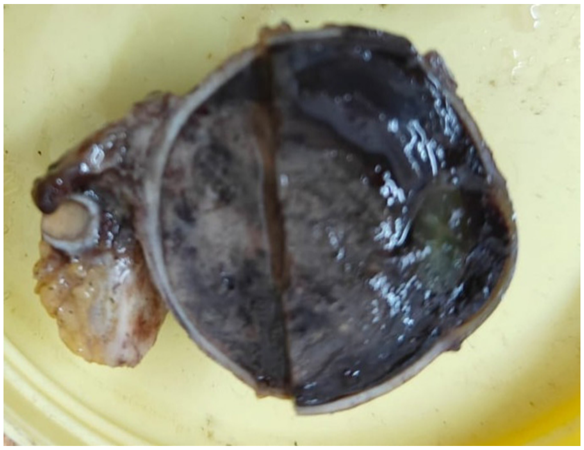

Figure 4. Gross photograph of the enucleated globe showing a dome-shaped, pigmented intraocular tumor in the choroid, with a basal diameter of approximately 1.9 cm.

Microscopy revealed spindle-shaped and oval melanocytic tumor cells arranged in interlacing fascicles within the choroidal layer. The cells showed moderate pleomorphism, nuclear hyperchromasia, and intracytoplasmic pigment granules. Melanophages, pigment-laden cells, and foci of necrosis were present (Figure 5). No mitotic figures were identified per 50 high-power fields, and tumoral necrosis was estimated at less than 5%. No scleral invasion, extrascleral extension, or optic nerve invasion was identified, and the tumor was classified as AJCC pT3a based on its dimensions. The overlying retina showed mild degenerative changes, and no retinal invasion by tumor cells was observed.

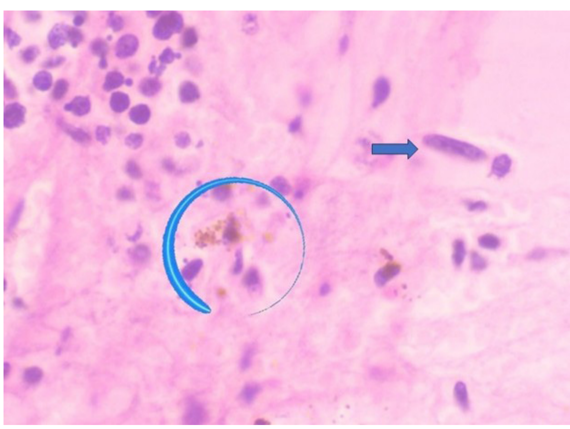

Figure 5. Histopathological section showing spindle-shaped melanocytic tumor cells (arrow) with pigment-laden cytoplasm and melanophages (circle).

Outcome and follow-up

The postoperative recovery was uneventful. Initial systemic evaluation, including abdominal ultrasonography, chest radiography, and liver function tests, revealed no evidence of distant metastasis. The patient was referred to the oncology department for risk stratification and long-term surveillance planning. Given the tumor size and histopathological features, structured metastatic surveillance with periodic hepatic imaging and liver function testing was recommended, along with chest imaging as clinically indicated. A three-month follow-up was scheduled, and the patient was advised to report any new symptoms, such as abdominal pain, weight loss, or jaundice, promptly.

Discussion:

Choroidal melanoma constitutes the majority of uveal melanomas, accounting for nearly 85–90% of cases [3,4]. Despite being the most common primary intraocular malignancy in adults, it remains rare in Asian populations. Greater ocular and cutaneous pigmentation, along with genetic and phenotypic differences, may contribute to the lower observed incidence in these groups [4-6]. Recognized risk factors include fair complexion, light iris color, germline or somatic BAP1 alterations, and cytogenetic abnormalities such as monosomy 3 and chromosome 8q gain [10,15]. In contrast to cutaneous melanoma, the role of ultraviolet radiation in the pathogenesis of uveal melanoma remains controversial, with epidemiological studies failing to establish a consistent causal link [5,6].

Clinical presentation is often nonspecific. Patients may notice painless visual decline, blurred vision, photopsia, or visual-field defects. On fundus examination, the lesion is classically seen as a pigmented dome-shaped or mushroom-shaped subretinal mass; exudative retinal detachment may accompany larger tumors [11]. In the present case, the patient had progressive painless loss of vision and a large pigmented dome-shaped lesion involving the posterior segment, matching these typical clinical features.

Multimodal imaging plays a central role in diagnosis. B-scan ultrasonography often shows a solid choroidal mass with low-to-medium internal reflectivity, acoustic hollowness, and, in some cases, choroidal excavation. MRI is useful for defining the intraocular extent and for assessing extrascleral spread. Melanin shortens T1 relaxation and alters T2 signal behavior, producing the characteristic pattern of T1 hyperintensity and T2 hypointensity seen in melanotic tumors [8,9]. These radiological findings also help differentiate choroidal melanoma from choroidal hemangioma, metastasis, and nevus [11].

On histopathology, uveal melanomas are broadly categorized as spindle-cell, epithelioid-cell, or mixed tumors. Spindle-cell morphology generally carries a more favorable prognosis than epithelioid-cell morphology, which is associated with a greater risk of metastasis [14]. Molecular features further refine prognostication; monosomy 3 and BAP1 mutation are among the strongest predictors of hepatic metastasis and poorer outcomes [10,15,16]. Other cytogenetic changes, including chromosome 8q gain and chromosome 1p loss, have also been linked to tumor progression and reduced disease-free survival.

Treatment selection depends on tumor size, location, visual potential, and the patient's systemic condition. Eye-conserving options, such as plaque brachytherapy, proton beam radiotherapy, and transpupillary thermotherapy, may be suitable for selected small and medium tumors [12,13]. Large tumors, particularly those exceeding approximately 10 mm in thickness or 16 mm in basal diameter, often require enucleation, especially when useful vision is unlikely to be preserved [12].

In metastatic or unresectable disease, immune checkpoint inhibitors and T-cell-redirecting therapies, including tebentafusp for HLA-A*02:01-positive patients, have demonstrated improved overall survival in recent trials, though outcomes remain suboptimal [17,18].

Tumor size remains one of the most important predictors of metastatic risk and survival. In the Collaborative Ocular Melanoma Study, large choroidal melanomas were associated with a substantial rate of histologically confirmed metastasis, most commonly involving the liver [6]. Metastasis may occur years after local tumor control, making long-term surveillance essential. Follow-up usually includes periodic hepatic imaging with ultrasonography, CT, or MRI, together with serial liver-function testing [17,18].

In this patient, the tumor was large but confined to the globe at the time of diagnosis, with no evidence of extrascleral spread or systemic metastasis on initial evaluation. The spindle-cell pattern is a relatively favorable histological feature; however, the large size of the tumor warrants careful long-term surveillance, particularly for delayed hepatic metastasis.

Conclusion:

Choroidal melanoma is uncommon in Asian populations but remains an important differential diagnosis in patients with unexplained progressive visual loss and a pigmented intraocular mass. This case highlights the need for a systematic diagnostic approach using fundus examination, ultrasonography, MRI and histopathological confirmation when surgical tissue is available.

MRI, particularly T1- and T2-weighted sequences, contributes valuable information on tumor morphology, internal signal characteristics and possible extraocular extension. Histopathology confirms the diagnosis and provides prognostic information based on tumor cell type and, where available, molecular profiling.

For large tumors with poor visual potential, enucleation remains an appropriate treatment option. Local control does not eliminate the risk of delayed systemic metastasis, especially to the liver. Lifelong follow-up with hepatic imaging and liver-function testing should therefore be emphasized. Timely diagnosis, patient counselling and coordinated ophthalmology-oncology care are essential to improve survival and quality of life in patients with this aggressive intraocular malignancy.

Declarations:

Declaration of patient consent:

The authors certify that all appropriate patient-consent forms have been obtained. The patient provided consent for reporting his clinical information and images in the journal. The patient understands that his name and initials will not be published and that reasonable efforts will be made to conceal his identity; however, complete anonymity cannot be guaranteed.

Conflicts of interest:

The authors declare no conflicts of interest related to this case report.

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